

ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

Implementation of the TensorFlow Model

The final model was developed to overcome the limitations observed in baseline models, particularly in handling contextual understanding and improving performance for neutral sentiment classification. A deep learning approach using TensorFlow was adopted to enhance overall accuracy and robustness.

Data Preparation

The dataset consists of approximately 35,000 educational feedback comments categorized into three sentiment classes: Negative, Neutral, and Positive.

- Labels were encoded as:
 - Negative → 0
 - Neutral → 1
 - Positive → 2
- The dataset was split into training and testing sets to ensure proper evaluation and generalization.

Text Processing

To convert raw text into a machine-readable format, a **TextVectorization layer** was used.

- Vocabulary size: 15,000
- Maximum sequence length: 120

This layer transforms textual data into numerical sequences, enabling efficient processing by the neural network.

Model Architecture

The final model was implemented using TensorFlow with the Keras API. A **Bidirectional LSTM (BiLSTM)** architecture was used to capture contextual dependencies in text.

Architecture Details:

- Embedding Layer → Converts words into dense vector representations
- Bidirectional LSTM → Captures context from both past and future sequences
- Dense Layer → Learns higher-level features
- Dropout Layer → Prevents overfitting
- Output Layer (Softmax) → Multi-class classification

This architecture allows the model to understand sentence structure more effectively compared to traditional models.

Model Training

- Optimizer: Adam
- Loss Function: Sparse Categorical Crossentropy
- Batch Size: 64
- Epochs: 15–20
- Callback Used: EarlyStopping (to avoid overfitting and restore best weights)

The training process was monitored using validation data to ensure stable learning.

Model Evaluation

The model was evaluated using standard classification metrics:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Final Performance:

- **Accuracy: ~82%**

Classification Performance:

- Negative → Strong precision and recall
- Neutral → Significant improvement compared to baseline
- Positive → High recall and stable performance

Confusion Matrix Insights:

- Most predictions lie along the diagonal, indicating correct classification
- Minimal confusion between classes compared to baseline
- Neutral class performance improved significantly

Comparison with Baseline

Metric	Baseline Model	Final Model
Accuracy	~77%	~82%
Neutral Recall	Low	Improved
Context Understanding	Limited	Strong

The final model clearly outperforms the baseline, especially in handling neutral sentiment and contextual relationships.

Inference Pipeline

The final system processes new input as follows:

1. Input comment
2. TextVectorization
3. Model prediction
4. Output label (Negative / Neutral / Positive)

This ensures real-time sentiment prediction capability.

Conclusion

The TensorFlow-based BiLSTM model significantly improves sentiment classification performance over baseline approaches. By leveraging deep learning and contextual sequence modeling, the system achieves higher accuracy and balanced predictions across all sentiment classes.

The model is scalable, efficient, and suitable for real-world deployment in educational feedback analysis systems.

RESULT

The final deep learning model was evaluated using classification metrics to measure its performance across all sentiment classes.

Sentiment Classification (BiLSTM Model)

The model achieved an overall accuracy of approximately 82%, showing a clear improvement over the baseline model.

Classification Report

Class	Precision	Recall	F1-Score	Support
Negative	0.80	0.89	0.84	2289
Neutral	0.82	0.84	0.83	2406
Positive	0.86	0.75	0.80	2389

Overall Accuracy: 0.82

Macro Avg F1: 0.82

Weighted Avg F1: 0.82

The model demonstrates balanced performance across all three sentiment classes, with significant improvement in the neutral class compared to the baseline model.

Confusion Matrix Analysis

The confusion matrix shows that most predictions are concentrated along the diagonal, indicating a high number of correct classifications.

- Negative class is well identified with high recall
- Neutral class performance is significantly improved
- Positive class shows strong precision with minor misclassification

Compared to the baseline model, the confusion between classes is reduced, especially for neutral sentiment.

Conclusion

The TensorFlow-based BiLSTM model outperforms the baseline model in terms of accuracy and class-wise performance. It effectively captures contextual relationships in text, resulting in improved sentiment classification. The model achieves stable and reliable performance, making it suitable for real-world applications.